

Adroit Technologies

Protocol Driver

Name	Omron Ethernet FINS Ethernet UDP/IP
DLL	OMR_ETH



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Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

<https://adroittech.co.za/>

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1. Overview

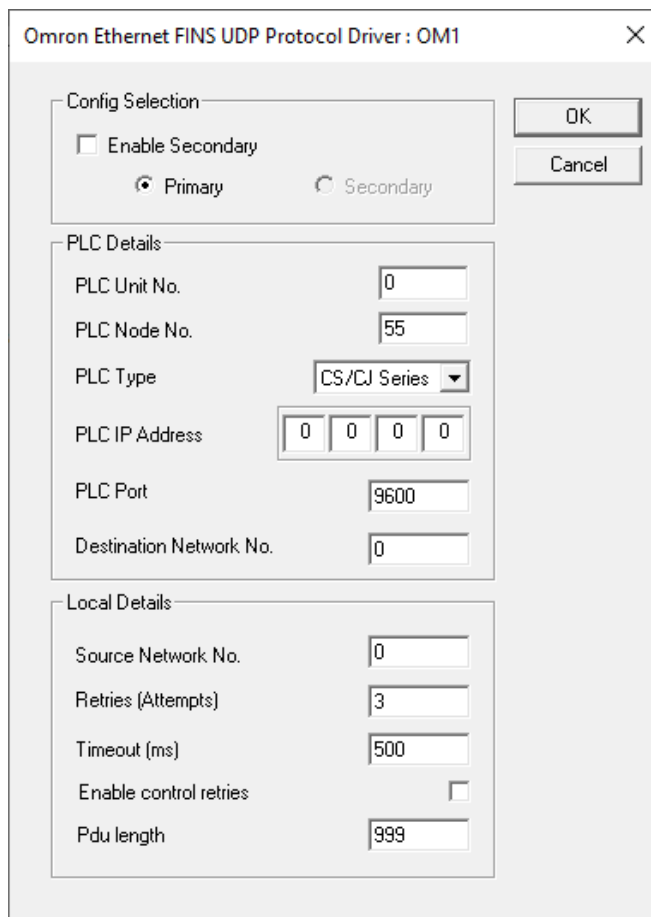
Adroit supports communication with the following PLCs: OMRON CS/CJ Series.

2. Wiring

Standard Ethernet interconnection (10BASE-T/E).

3. Device Configuration

3.1. Device Configuration



3.1.1. Config Selection

Enable Secondary – Enables secondary configuration for dual redundant PLCs via **Primary** and **Secondary** radio buttons. Do not enable if a secondary channel is not configured correctly.

3.1.2. PLC Details

PLC Unit no. – The unit number as configured at the front of the Ethernet unit.

PLC Node no. – The node number as configured at the front of the Ethernet unit. The node number is the same as the last byte of the IP Address. For example, if the IP Address is 192.168.1.55, then the node number should be 55.

PLC Type – The CS/CJ Series is currently supported.



PLEASE NOTE

If needed, the CV series will be accommodated.

The following PLC types can be chosen:

PLC Type	Max number of read elements (p/s)	Max number of write elements (p/s)
OMRON CS/CJ Series	999 words	996 words
OMRON CV Series*	999 words	996 words

* Currently not available

PLC IP Address – The IP Address of the PLC as configured at the back of the OMRON Ethernet unit.

PLC Port – The protocol currently used in the OMRON Ethernet driver is UDP, which always operates on port 9600. This is set by default.

Destination Network No. – Change the FINS network number to correspond to the network the PLC is assigned to.

3.1.3. CPU Unit Data / Status (currently unavailable)

By pressing the RETRIEVE button, the settings as provided in the previous section ([PLC Details](#)) are tested on the actual network. Two packets are sent out, of which the first checks the current CPU mode and the second checks the CPU Unit model and version.

The four possible responses are:

Status	Model	Version	Interpretation
RUN mode!	For ex. CS1G_CPU45H	For ex. 2.30	CPU is functioning correctly and in RUN mode.
STOP/Program mode!	For ex. CS1G_CPU45H	For ex. 2.30	CPU is currently in STOP / Program mode
MONITOR mode!	For ex. CS1G_CPU45H	For ex. 2.30	CPU is in monitor mode (a type of RUN mode)
Not connected!	Unknown	Unknown	CPU connection couldn't be made

3.1.4. Local Details

Source Network No. – Change the FINS network number to correspond to the network the ADROIT PC is assigned to. 0 if only one segment, but if multiple segments change it to the last segment PLCs' network number.

Retry – The number of times the driver must retry a connection, or a Transmit/Receive transaction before returning an indication of failure.

Timeout – The time in milliseconds to wait between each communication retry.

Enable control retries – By checking this box, an interrupted WRITE operation will be retried. Otherwise, only READ operations will be retries.

3.2. Registry Settings (BYTE, WORD and DWORD Swopping)

There are a few registry settings that allows for compensation when bytes, words or double words get swapped around in the receiving protocol message for the different post fixes (as shown below) types of namely B, D, I, L, N, R, W and X.

These registry keys are not accessible from the dialog but can be changed if required from the registry by the following steps:

1. Open the Registry by going to START-RUN and typing regedit and then press ENTER
2. Go to the following key –
HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\Omr_Eth\<YOUR DEVICE NAME>\Primary
3. There is a set of DWORD Keys that can be set to 0 as it is 1 by default.

The keys are:

"Swap_B_Bytes"	– Swop the bytes of a 16-bit BCD value
"Swap_D_Bytes"	– Swop the bytes of an unsigned double value
"Swap_D_Words"	– Swop the words of an unsigned double value
"Swap_I_Bytes"	– Swop the bytes of a signed word value
"Swap_W_Bytes"	– Swop the bytes of an unsigned word value
"Swap_L_Bytes"	– Swop the bytes of a signed double value
"Swap_L_Words"	– Swop the words of a signed double value
"Swap_N_Bytes"	– Swop the bytes of a 32-bit BCD value
"Swap_N_Words"	– Swop the words of a 32-bit BCD value
"Swap_R_Bytes"	– Swop the bytes of a 32-bit Float value
"Swap_R_Words"	– Swop the words of a 32-bit Float value
"Swap_X_Bytes"	– Swop the bytes of a 64-bit Float value
"Swap_X_Words"	– Swop the words of a 64-bit Float value
"Swap_X_DWords"	– Swop the double words of a 64-bit Float value

4. Supported Addresses

OMRON CS/CJ PLC data address types supported by the OMRON Ethernet Adroit Protocol driver

Area	Code	Min Range	Max Range	PLC Data Type	Boolean Addressing Example	Integer Addressing Example	Real Addressing Example	String Addressing Example
CIO area	CIO	0000.00	614315	Bit	CIO0000.15			
	CIO	0000	6143	Word		CIO0001	CIO0001	
Work area	WR	000.01	511.15	Bit	W00001			
	WR	000	511	Word	-	W001	W001	
	WR	000	511	DWord		W001 D	W001 D	
Holding bit area	H	000.00	511.15	Bit	H510.15			
	H	000	511	Word		H511	H511	
Auxiliary bit area	AR	000.00	447.15 (Read-only)	Bit	A447.15			
	AR	000	447 (Read-only)	Word		A447	A447	
	AR	448.00	959.15 (Read/Write)	Bit	A959.15			
		448	959 (Read/Write)	Word		A959	A959	
Timer area	TIM	0000	4095	Bit	T0000.00			
	TIM	0000	4095	Word		T1234	T1234	
Counter area	CNT	0000	4095	Bit	C0000.00			
	CNT	0000		Word		C4095	C4095	
DM Area	DM	0000.00	32767.15	Bit	D12345.15			
	DM	00000	32767	Word		D11224	D11224	D11224 S20
EM Area	EM	0000.00	32767.15	Bit	E12345.15			
	EM	00000	32767	Word		E11224	E11224	E11224 S22

Note 1: Only certain addresses of the Auxiliary Bit area are writeable as well. Note which areas are read-only.

Adding any of the following postfix characters onto the address will format the data accordingly:

B = 16-bit BCD
N = 32-bit BCD
W = Unsigned WORD
I = Signed WORD
D = Unsigned DOUBLE
L = Signed Double
R = 32-bit FLOAT
X = 64-bit FLOAT
S = String support

5. Protocol Definition and Considerations

5.1. Advanced Options

5.1.1. Scanning fast as possible is supported in this driver

Any Agent Server with at least rev34 of t_scan.dll, will provide the following registry key:
 HKEY_LOCAL_MACHINE\SOFTWARE\AdroitTechnologies\Adroit\Scanning\AllowFastAsPossible

Set this to true, AND scan any required addresses at 1msec, the system will then attempt to scan the scan tasks that contain those addresses as fast as possible. Warning: Keep an eye cpu load so as not to cause the system to become unstable.

5.2. Message Protocol

5.2.1. FINS Header Structure (Prefix structure for ALL messages)

Note: There is no CRC field present in the FINS data frame, although the response codes as indicated in this section provide minimal assurance that the correct data was received.

FINS packet header: (Note that the numbers are in hex unless otherwise noted)

ICF	RSV	GCT	DNA	DA1	DA2	SNA	SA1	SA2	SID
(Typically 80 or C0 hex)	00	07	00	37	00	00	11	00	01

ICF:

* The bits for the ICF are laid out as follows:

Bit 7: (1)	Bit 6: (0)	Bit 5: (0)	Bit 4: (0)	Bit 3: (0)	Bit 2: (0)	Bit 1: (0)	Bit 0: (1)
------------	------------	------------	------------	------------	------------	------------	------------

Bit 0: Response (0=Required, 1=Not required) ,

Bit 1: Always 0

Bit 2: Always 0

Bit 3: Always 0

Bit 4: Always 0

Bit 5: Always 0

Bit 6: Data classification (0=Command, 1=Response),

Bit 7: Bridges (0=Not used, 1=used).

A typical ICF for a command frame would be 0x80, and for a response frame 0xC0.

RSV:

Reserved byte. Always 0.

GCT:

Gateway count. Number of bridges passed through. Currently, the Adroit OMRON Ethernet protocol driver doesn't support bridging, thus this byte will be 0x00.

DA1:

Destination node address. This corresponds to the last byte of the destination IP Address, which, in this example is 55 (HEX 0x37)

DA2:

Destination unit address. Corresponds to the unit number as set on the destination unit.

SNA:

Source network address. 0x00 indicates that it is the local network.

SA1:

Source node address.

SA2:

Source unit address.

SID:

Service ID. This is used to identify the process generating the transmission. The SID can be set to any number between 0x00 and 0xFF.

FINS command bytes (The following bytes are appended to the FINS header to provide memory read and functionality):

5.2.2. FINS Memory Area Read Function

Command Format:

MRC	SRC	I/O Memory area code*	Beginning address	No. of items
01	01	xx	000000	0001

Response Format:

MRC	SRC	End code 1	End code 2	Data (for number of items)
01	01	00	00	0000xxxxxx

* A table of the different I/O memory area codes are given after the layout of the Memory Area Write function.

MRC:

Main request code.

SRC:

Sub-request code.

I/O Memory code:

The memory code for the specific area and data type you intend to read/write. See the table provided after the description of the Memory Area Write function for a layout of the available codes.

Beginning address:

The PLC address where reading/writing will start.

No. of items:

Indicates the number of items (whether it be bits or words), to read or write.

End code 1 and 2:

The following table provides a brief overview of the most important end codes that may be received:

End code 1	End code 2	Meaning
00	00	Normal completion
01	01	Local node not in network
01	05	Node address range error
02	01	Destination node not in network
02	04	Destination node busy
03	02	CPU Unit error
03	04	Unit number error
10	01	Command too long
10	02	Command too short
10	03	Elements / data do not match
10	04	Command format error
10	05	Header error
20	02	Read not possible : Protected
21	01	Write not possible : Read-only
21	02	Write not possible : Protected
22	03	Wrong PLC mode : PLC is in program mode
22	04	Wrong PLC mode : PLC is in DEBUG mode
22	05	Wrong PLC mode : PLC is in MONITOR mode

NOTE: In addition to the above codes, there are also three flags provided in the end codes, which is bit 6,7 and 15. If bit 15 is ON, an error has occurred in a network relay operation. If bit 6 or 7 is ON an error has occurred in the destination CPU unit.

5.2.3. FINS Memory Area Write Function

Command Format:

MRC	SRC	*I/O Memory area	Beginning address	No. of items	Data (for number of items)
01	02	xx	000000	0001	0000xxxx

Response Format:

MRC	SRC	End code 1	End code 2
01	01	00	00

5.2.4. I/O Memory Area Codes

Area	Data type	Code
CIO	Bit	30
CIO	Word	B0
WR	Bit	31
WR	Word	B1
HR	Bit	32
HR	Word	B2
AR	Bit	33
AR	Word	B3
Timer	Flag/Bit	09
Timer	Word	89
Counter	Flag/Bit	09
Counter	Word	89
DM	Bit	02
DM	Word	82
EM	Bit	82
EM	Word	82

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

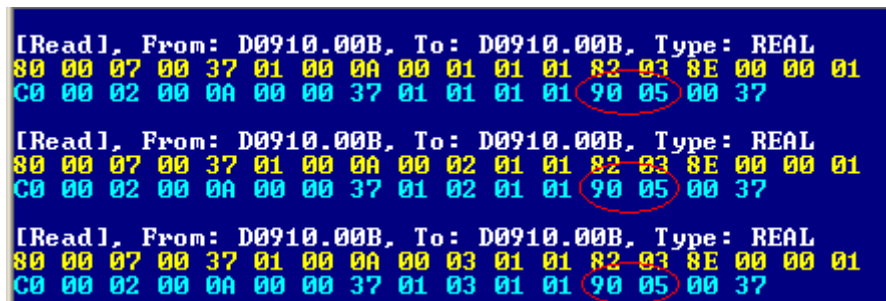
6.3.1. Problem 1

Problem

When connected to the PLC and there is communication with device, but my Adroit Tag is still scan bad even though I can see in the datascope that I am receiving some sort of data?

Solution

The first place to look at is in the datascope on ASCII mode. Bytes 13 and 14 of the response message will indicate the END CODES. If the 2 bytes are not 00 00 then the tag is "scan bad" as there is some kind of error as indicated on image below.



```

[Read], From: D0910.00B, To: D0910.00B, Type: REAL
80 00 07 00 37 01 00 0A 00 01 01 01 82-03 8E 00 00 01
C0 00 02 00 0A 00 00 37 01 01 01 01 90 05 00 37

[Read], From: D0910.00B, To: D0910.00B, Type: REAL
80 00 07 00 37 01 00 0A 00 02 01 01 82-03 8E 00 00 01
C0 00 02 00 0A 00 00 37 01 02 01 01 90 05 00 37

[Read], From: D0910.00B, To: D0910.00B, Type: REAL
80 00 07 00 37 01 00 0A 00 03 01 01 82-03 8E 00 00 01
C0 00 02 00 0A 00 00 37 01 03 01 01 90 05 00 37
  
```

Above Image Represents ASCII datascope with end codes returned an error.



```

^TX: [Read], From: D0910.00B, To: D0910.00B, Type: INTEGER
^RX: 80 00 07 00 37 00 00 50 00 01 01 01 82-03 8E 00 00 01
^TX: C0 00 02 00 50 00 00 37 00 01 01 01 00 00 00 30
^RX: [Read], From: D0911.00B, To: D0912.00B, Type: REAL
^TX: 80 00 07 00 37 00 00 50 00 01 01 01 82-03 8F 00 00 02
^RX: C0 00 02 00 50 00 00 37 00 01 01 01 00 00 00 50 00 14
^TX: [Read], From: D0910.00B, To: D0910.00B, Type: INTEGER
^RX: 80 00 07 00 37 00 00 50 00 02 01 01 82 03 8E 00 00 01
^TX: C0 00 02 00 50 00 00 37 00 02 01 01 00 00 00 30
  
```

Above Image Represents ASCII datascope with no errors.

The error codes are available in this document on page 7, but the OMRON Communication Commands Reference Manual – Catalogue Number W342-E1-06 goes into more depth with the error codes on page 113 – 119.

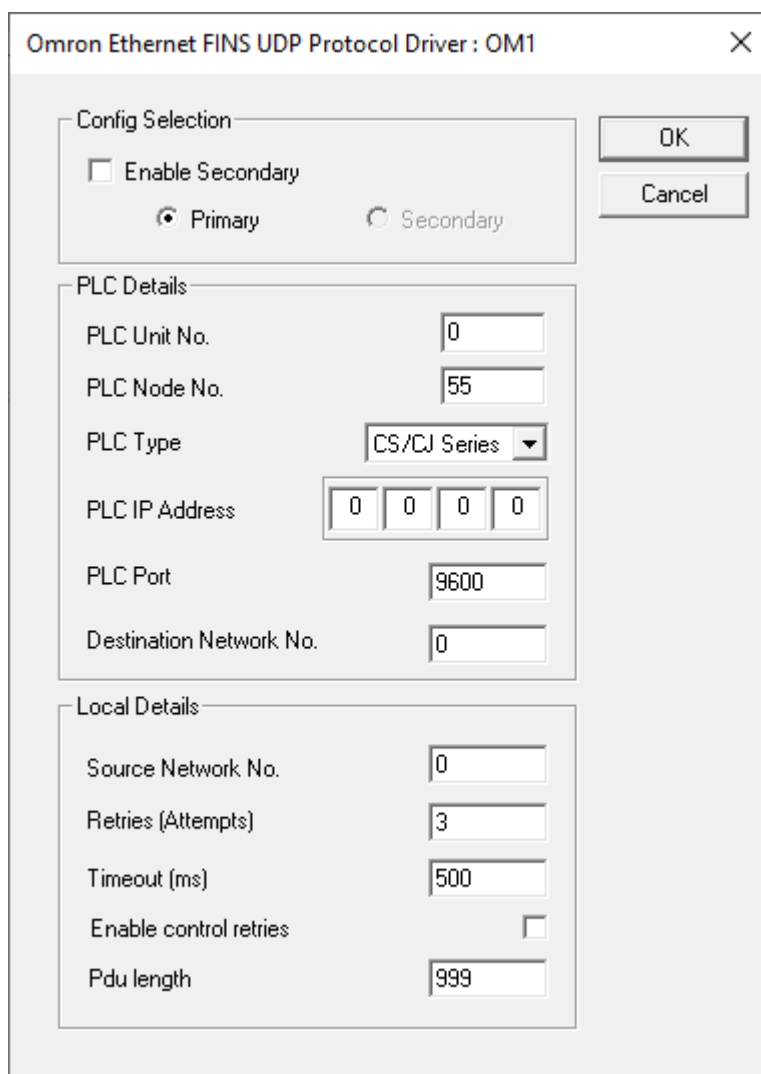
Example: End Code = 90 05 hex.

Take 9005 and convert to binary, which will give you 10001100101101. Stripping of the MSB (Bit 15) and in this case was ON which means "Error in Network relay" as shown on pg118 of above mentioned OMRON manual. After stripping bit 15 we are left with 0001100101101. Bit 8 to 14 represent the main end code which will give us value of 10. In the OMRON manual as mentioned above on page 114 the value 10 means "Command Format Error".

If bit 7 is ON it means a "Fatal CPU unit error" and with bit 6 ON it means "Non-fatal CPU Unit Error". In our case we did not have a CPU error.

Bit 0 to 5 represents the sub code which in our case is 05. According to the OMRON manual on page 114 the sub end code of 5 means "Header Error".

This concludes that our errors received is "error in network relay", "command format error" and "header error" and the problem was solved by correcting the "Unit No" that was specified incorrectly in the adroit setup as indicated below the Unit No should have been 0.



Omron Ethernet FINS UDP Protocol Driver : OM1

Config Selection

☐ Enable Secondary

☒ Primary ☐ Secondary

PLC Details

PLC Unit No.

PLC Node No.

PLC Type

PLC IP Address

PLC Port

Destination Network No.

Local Details

Source Network No.

Retries (Attempts)

Timeout (ms)

Enable control retries ☐

Pdu length

OK Cancel

6.3.2. Problem 2

Problem

We can write out to a 16-bit BCD but not read anything back. It just shows 0.

Solution

We could not reproduce it on our PLC in the office, and then eventually found that when you go to CX Programmer V5.00, connect online and click on "PLC – Clear all memory areas" menu does resolve the issue.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	09-Jan-2023	K. Robinson	Grammar and spelling edits, alignment and general editing.	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
	06-Jan-2005	First release notes! Beta Release Available
	07-Sep-2006	Added suffix explanation and troubleshooting to document
19	19-Sep-2012	Added Extended Memory Support as of Revision 19
	10-Oct-2012	Added Registry settings not available from the dialog and added the suffix R into document!
	22-Nov-2013	Improved timing and speed capability of this driver.
	06-Aug-2018	String support documented; I/O ranges fixed.
27	03-Oct-2022	Datascope cleaned up. Fixed confusing addressing where chars like "F" were allowed but "I" functionality implemented.

